

EcoBite AI — Smart Refrigerator Food Waste Reduction System

An AI-powered smart refrigerator assistant designed to reduce household food waste through real-time inventory tracking, expiry prediction, and intelligent alerts.

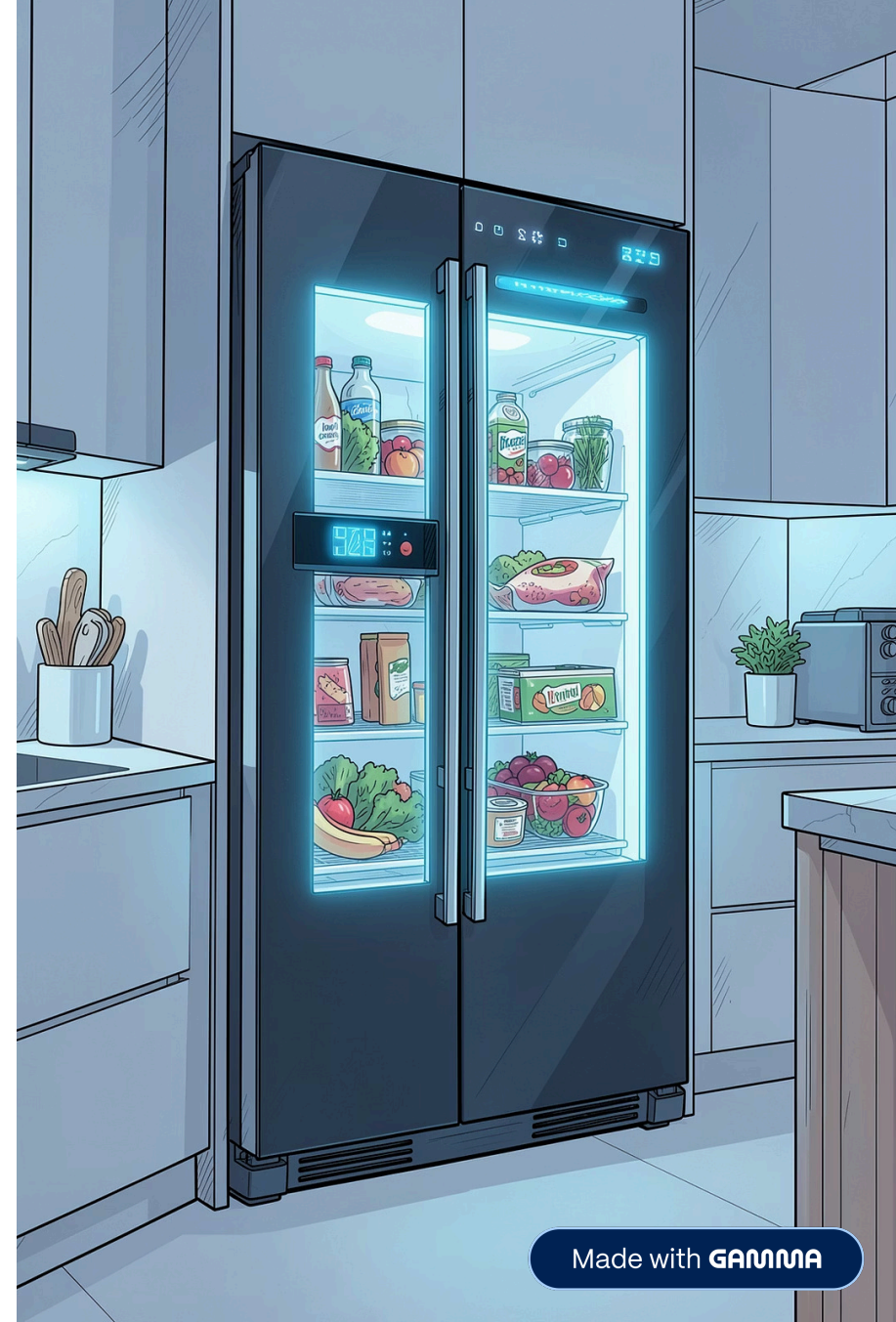
1M1B AI FOR SUSTAINABILITY VIRTUAL INTERNSHIP

IBM SKILLSBUILD × AICTE

SDG 12 — RESPONSIBLE CONSUMPTION

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THE PROBLEM

Food Waste Is a Silent Crisis

Families discard edible food daily — forgetting stored items or missing expiry dates. This causes financial loss and greenhouse gas emissions from decomposing food in landfills.

Forgotten Food

Items pushed to the back are forgotten until they spoil.

No Smart Alerts

No affordable system proactively alerts users before food becomes unusable.

Financial Loss

Wasted food means wasted money — a burden for middle and low-income households.

Environmental Damage

Food in landfills produces methane — a greenhouse gas far more potent than CO₂.

WHO IS AFFECTED

Every Household Faces This Challenge



Middle & Low-Income Families

Face real financial loss from wasted groceries.



Students & Hostel Residents

Irregular schedules make systematic fridge checks difficult.



Working Professionals

Busy lifestyles leave little time to track fridge contents.



Elderly Individuals

May struggle to track stored food or read small expiry labels.

Supporting the United Nations Sustainable Development Goals

EcoBite AI directly supports **SDG 12, Target 12.3** — halving per-capita global food waste by 2030 — turning everyday refrigerator usage into an act of sustainability.



SDG 12 — Primary
Responsible Consumption & Production — Reduces household food waste and builds sustainable management habits.



SDG 2 — Secondary
Zero Hunger — Preserves edible food for consumption or donation, improving food security.



SDG 11 — Secondary
Sustainable Cities — Reduces landfill burden and supports cleaner urban environments.



SDG 13 — Secondary
Climate Action — Less food waste means reduced methane emissions from landfills.

Introducing EcoBite AI

EcoBite AI integrates an AI-powered smart refrigerator with a companion mobile app via IoT connectivity — transforming a passive appliance into an active food management partner.

Core Capabilities

- Real-time food inventory tracking via mobile app
- AI-powered expiry prediction and consumption priority ranking
- Two-stage smart notification system (7-day + critical alerts)
- Physical LED compartment alerts for critical expiry
- AI-generated recipe suggestions and food donation recommendations



HOW IT WORKS

Five-Stage Intelligent Workflow

From food entry to AI recommendations — EcoBite AI operates across five intelligent stages, ensuring no food item is ever forgotten or wasted.



Food Registration

User registers items via app — manual entry or barcode/QR scan, recording name, quantity, expiry, and compartment location.



Cloud Sync via IoT

Smart refrigerator syncs all food data with the mobile app in real time for live inventory access.



AI Monitoring

AI analyses inventory, assigns consumption priority scores, and detects high-risk items likely to be wasted.



Smart Notifications

Two-stage alerts: 7-day reminder, then 1–2 day critical alert with LED blinking on the specific compartment.



AI Recommendations

Recipe suggestions, food storage tips, donation recommendations, and personalised waste reduction insights.

Two-Stage Smart Notification System

EcoBite AI's dual alert system ensures users are informed both digitally and physically — making it impossible to miss a critical expiry warning.

First Alert — 7 Days Before Expiry

Mobile app sends a push notification:
"Your milk will expire in 7 days. Consider using it this week."
Gives users time to plan meals around the item.

Critical Alert — 1–2 Days Before Expiry

Urgent push notification + **LED compartment blinks red** — upper or lower section depending on item location. Impossible to miss even without phone.



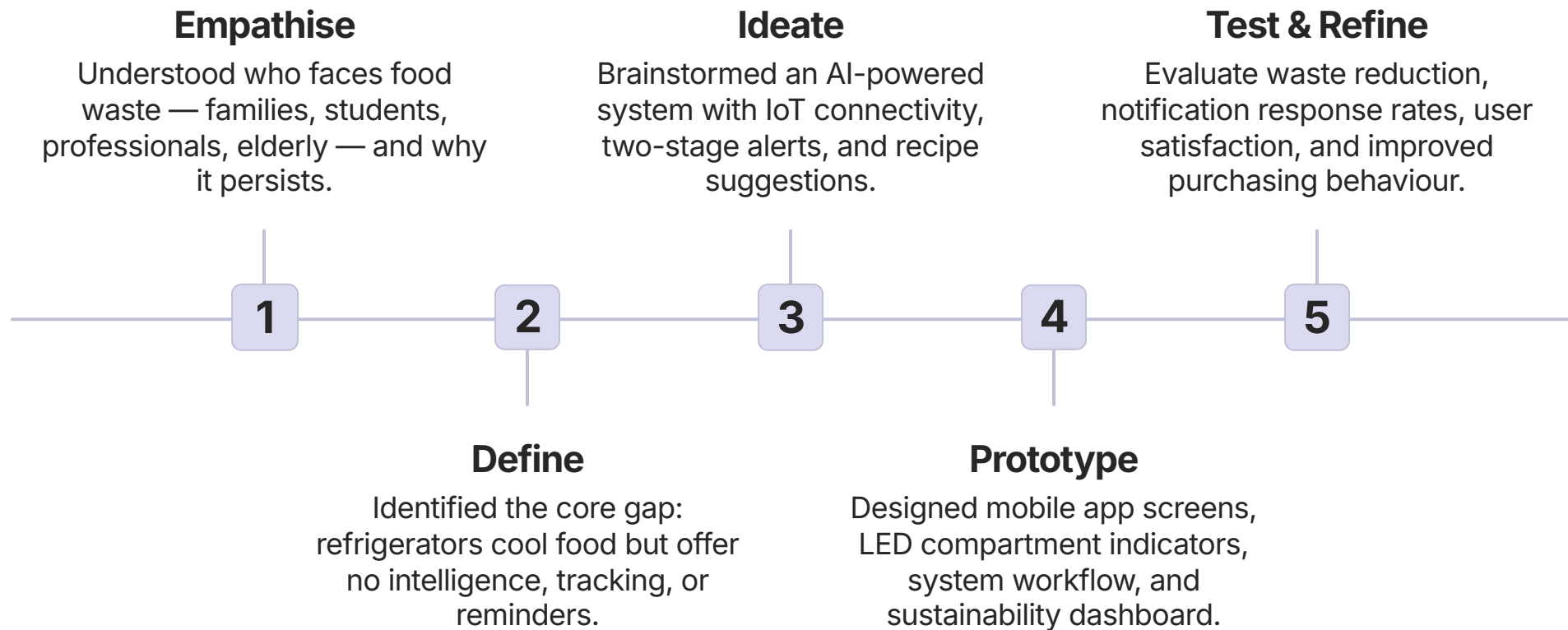
Why Artificial Intelligence?

Traditional refrigerators cool food but offer no intelligence. EcoBite AI uses AI to transform a passive appliance into an active food management partner.

AI Component	Application in EcoBite AI
Generative AI	Generates recipe suggestions and food storage tips based on available inventory.
Conversational AI	Chatbot answers questions like "What food is expiring soon?" or "What can I cook?"
Entity Extraction	Identifies food names, quantities, expiry dates, and compartment locations from inputs and scans.
Recommendation Engine	Suggests recipes, donation options, and storage improvements based on inventory and behaviour patterns.
AI Decision Support	Ranks items by consumption urgency and generates personalised waste reduction insights.
IoT Data Processing	Continuously processes inventory data from the refrigerator's sensors and LED controls.
Notification Automation	Triggers push notifications and LED alerts based on AI-calculated expiry risk scores.

Human-Centred Design Process

EcoBite AI was developed using the Design Thinking framework — ensuring the solution is grounded in real user needs.



Eat Smart. Waste Less. Live Sustainably.

Expected Impact at Scale

Social

Reduced food waste and improved nutrition planning for families.

Economic

Lower grocery expenses through better food utilisation and reduced spoilage.

Environmental

Reduced methane emissions and decreased solid waste at household level.

Community

Surplus food redirected to donation, improving local food security.

Responsible AI Principles

Fairness — Works equally for all users regardless of age or income.

Transparency — Clearly explains why each item is flagged as priority.

Privacy — Food data stored securely, never shared without consent.

Accessibility — Simple interface with large text and voice interaction.

□ EcoBite AI demonstrates that AI for sustainability, when applied thoughtfully, can create real-world impact — contributing to responsible consumption, zero hunger, and climate action.